

Soft Spaces - Phase 3.2

Working Note: Local Hotspot Mechanism

Plan for v32.3 after closure of the global +/-E symmetry chain

Status at start of this note. The global +/-E organization is no longer treated as an unexplained numerical pattern. For the frozen reference model, the chain $\Gamma H \Gamma^\dagger = -H \rightarrow$ paired eigenspaces $\rightarrow$ paired PVQ structure $\rightarrow \Sigma_{-E} = -\Gamma \Sigma_E \Gamma^\dagger$ has been verified numerically. The remaining scientific question is therefore local: why are some symmetry-related +/-E pairs strong hotspots while others are weak?

1. Purpose of the next step

The next program, `v32_3_hotspot_local_mechanism.py`, is not intended to rediscover the global symmetry and is not intended to launch a large new parameter scan. Its purpose is to search for a compact local diagnostic that varies from pair to pair and tracks the already frozen hotspot prominence.

Primary working question:

Which local P/Q quantity best explains the variation in hotspot prominence along an already symmetric +/-E spectrum?

2. What remains frozen

- 8-qubit reference model and the 11-term Hamiltonian construction.
- Reference seed 25042000 for the first diagnostic run.
- Degeneracy-closed P eigenspaces and exact +/-E pairing.
- The Phase 3.1 REAL/NULL construction and perturbation families: dephasing and transverse.
- The regularized second-order kernel $g_\eta(x) = x/(x^2 + \eta^2)$, with $\eta = 1e-3$.
- The existing Phase 3.1 prominence. v32.3 may correlate against it, but must not redefine it.

3. The two-level picture

The project now naturally separates into two mathematical layers. The first is global and already identified; the second is local and still open.

Global layer: $\Gamma H \Gamma^\dagger = -H$. This generates the exact $E \leftrightarrow -E$ pairing and constrains P, Q, PVQ and the second-order operator Σ .

Local layer: A yet-unidentified scalar or small set of scalars determines where the prominence becomes large. This is the target of v32.3.

4. Candidate local quantities

For every exact +/-E pair, and separately for dephasing and transverse perturbations, v32.3 should calculate and compare the following candidate descriptors:

- $||PVQ| |_{F^2}$: total perturbative coupling strength from P into Q.
- $||\text{Sigma}_E| |_F$: magnitude of the regularized second-order effective response.
- Delta_PQ: nearest spectral separation between the P energy and the complementary Q spectrum.
- $||PVQ| |_{F^2} / \text{Delta_PQ}$: a simple coupling-to-gap scale, used only as a diagnostic candidate.
- $||\text{Sigma}_E| |_F / \text{Delta_PQ}$: a second diagnostic ratio, again without treating it as a derived law.
- REAL-NULL differences for the same quantities, because the hotspot score itself was defined relative to NULL.
- Optional local contrast: the value at the pair relative to nearby eigenspace pairs, to test whether prominence is associated with a local extremum rather than with absolute magnitude.

5. Symmetry-aware reduction

Because v32.2 showed that the REAL +/-E partners have equal unitary-invariant norms to numerical precision, v32.3 should not treat +E and -E as two independent observations. For each pair, the REAL value can be represented by the mean of the two partners; the +/- asymmetry should still be retained as a quality-control field, but not counted twice in correlation statistics.

This avoids pseudo-replication and keeps the statistical unit equal to one exact +/-E pair.

6. Planned statistical diagnostics

The program should remain descriptive. It should not claim a predictive law from one seed. The first run should therefore report:

- Pearson correlation between frozen pair prominence and each candidate quantity.
- Spearman rank correlation, so a monotonic relation can be detected even if it is not linear.
- TOP / MIDDLE / BOTTOM prominence-group medians for each candidate.
- Maximum +/-E asymmetry as a symmetry quality check.
- A ranked list of candidate descriptors by absolute Spearman correlation.

7. Decision logic

Outcome A - one compact candidate dominates. If the same physically interpretable quantity is strongly associated with prominence in both perturbation families, it becomes the leading local hotspot mechanism candidate.

Outcome B - REAL-NULL contrast dominates. Then hotspot strength may be governed less by the absolute REAL coupling and more by how strongly REAL departs from the randomized control.

Outcome C - no simple scalar dominates. Then the next model should combine two or more descriptors, for example coupling strength and spectral denominator, rather than forcing a one-variable formula.

8. What v32.3 must not conclude

- A strong correlation on seed 25042000 is not yet a universal law.
- A ratio such as $||PVQ| |_F^2/\Delta_PQ$ is a diagnostic ansatz, not a theoretically established formula.
- The global Gamma symmetry explains +/-E pairing, but by itself it does not explain prominence.
- NULL comparisons must be kept distinct from REAL algebraic identities.

9. Expected outputs

The program should create a human-readable text report and a CSV table with one row per exact pair and perturbation family. The CSV should contain the frozen prominence and all candidate local descriptors so that later across-seed work can reuse the same schema.

10. Working equation for this stage

At this stage the project should deliberately keep the local law open. A useful schematic expression is:

Hotspot prominence $\sim F(||PVQ| |_F^2, ||\text{Sigma_E}| |_F, \Delta_PQ, \text{REAL-NULL contrast, local spectral context})$

The purpose of v32.3 is to learn which arguments of F matter. Only after that should Phase 3.2 attempt a simpler closed-form hotspot expression.

11. Immediate next action

Run `v32_3_hotspot_local_mechanism.py` on the frozen reference seed 25042000. Inspect the ranked correlation table and TOP/MIDDLE/BOTTOM contrasts. If a stable candidate emerges, the next version should test it across the frozen seed ensemble rather than immediately adding more model complexity.

Phase 3.2 status: global symmetry mechanism closed for the reference model; local hotspot-selection mechanism remains open.